THE UNIVERSITY OF AUCKLAND

SEMESTER TWO 2024
Campus: City

MECHANICAL ENGINEERING

Biomechatronic Systems

(Time Allowed: TWO hours)

NOTE:

1. This exam is worth 50% of your final grade.
2. It will be marked out of 100 marks.
3. There are 3 parts. Part 1 and 3 are worth 30 marks each and part 2 is worth 40 marks.
4. You must answer ALL questions.
5. Answer each question in the space provided below the question.
6. If you require additional answer pages, overflow pages are provided at the end of this booklet. Within the designated space for the question you answer, you must provide a clear indication that your answer is continued on an overflow page e.g., "Answer is continued on overflow page X".
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8. Cross out notes and other work you do not wish to have assessed.

Surname: ……..…….. Forenames: ……

University of Auckland ID number: ……

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PART 1: EEG and sleep (30 marks)

For each of the following multiple-choice questions (questions 1 to 20), circle one answer only. "REM" refers to "rapid eye movement". Each of these multiple-choice questions is worth 1 mark.

1. Which one of the following is not a sleep state or a sleep stage?

a) spindles
b) N2
c) non-REM
d) slow-wave sleep

2. Which one of the following statements about sleep is incorrect?

a) Memory deficits affect events occurring near sleep onset.
b) A sleep clinician should expect to see more slow-wave sleep in older men.
c) Insomnia and hypersomnia are sleep disorders.
d) In normal, young adults there is more non-REM sleep early in the sleep episode as compared to late in the sleep episode.

3. Which one of the following definitions best applies to "hypnogram"?

a) A graph that represents the spontaneous electrical activity of the brain.
b) A graph that represents the stages of sleep as a function of time.
c) A graph that represents the heart's electrical activity as patient transitions between sleep and wakefulness.
d) An electrical device used to gather and amplify a biological signal.

4. Which one of the following statements is incorrect?

a) The traditional terminology for non-REM sleep stages is: stage 1, stage 2, stage 3, stage 4.
b) The new terminology for non-REM sleep stages, introduced by the American Academy of Sleep Medicine in 2007, is: N1, N2, N3.
c) Human slow-wave sleep is synonymous with sleep stage N2.
d) Human slow-wave sleep is synonymous with sleep stage 3 and stage 4.

5. Which one of the following statements is incorrect?

a) Normal sleep begins in non-REM.
b) Normally, the first period of REM sleep occurs about 80 to 100 minutes after sleep onset.
c) REM sleep episodes shorten across the night.
d) The REM/non-REM sleep cycle is about 90 minutes in duration.

6. Which one of the following is the primary purpose of a continuous positive airway pressure (CPAP) machine?

a) To provide medication for insomnia.
b) To improve sleep quality through sound therapy.
c) To monitor sleep patterns.
d) To keep the airway open during sleep.

7. Which one of the following electroencephalogram (EEG) patterns is not characteristic of sleep?

a) high-amplitude slow waves
b) alpha rhythms
c) spindles
d) K-complexes

8. Which one of the following provides a "shorthand" definition of non-REM sleep?

a) A relatively active brain in a movable body.
b) A relatively inactive brain in a paralyzed body.
c) A relatively inactive brain in a movable body.
d) A relatively active brain in a paralyzed body.

9. Regarding sleep onset, which one of the following statements is incorrect?

a) The electromyogram (EMG) may show gradual diminution of muscle tone as sleep approaches.
b) The electrooculogram (EOG) may show slow eye movements as sleep approaches.
c) As sleep approaches, the electroencephalogram (EEG) changes from a low-voltage, mixed-frequency pattern to clear, rhythmic alpha activity.
d) The brain is unresponsive to visual input immediately after sleep onset.

10. Which of the following statements about healthy adult sleep is correct?

a) All of these.
b) The sleep of young adult males shows no difference to that of young adult females.
c) The transition from wake to sleep produces a memory impairment.
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11. Which of the following statements about sleep arousal threshold is correct?

a) A person likely has a lower sleep arousal threshold when hearing their own name versus someone else's name.
b) Stage 1 sleep is associated with a high arousal threshold.
c) Children cannot be roused from REM sleep.
d) Low arousal threshold is typical of deep sleep.

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12. Which one of the following statements is incorrect?

a) Sleep is entered through non-REM sleep.
b) Slow-wave sleep predominates in the last third, and REM sleep predominates in the first third of the night.
c) REM and non-REM sleep alternate with a period near 90 minutes.
d) Wakefulness usually accounts for less than 5% of the night.

13. Which one of the following statements is correct?

a) Stage N2 sleep constitutes about 13% to 23% of sleep.
b) Stage N3 sleep constitutes about 45% to 55% of sleep.
c) REM sleep constitutes about 12% to 14% of sleep.
d) Stage N1 sleep constitutes about 2% to 5% of sleep.

14. For people aged 30 to 70 years, what is the prevalence of obstructive sleep apnea?

a) 1%
b) 25%
c) 0.5%
d) Between 70 and 80%

15. Which one of the following statements about the apnea-hypopnea index (AHI) is incorrect?

a) The AHI is commonly used to define obstructive sleep apnea, and its severity.
b) The AHI is the average number of apnoea and hypopnea events (lasting > 10 sec) that occur every hour.
c) AHI < 5 indicates mild obstructive sleep apnea.
d) AHI > 30 indicates severe obstructive sleep apnea.

16. Which of the following is not a risk factor for obstructive sleep apnea?

a) Dehydration
b) High blood pressure (hypertension)
c) Smoking
d) Old age

17. Which one of the following is not a sign/symptom of obstructive sleep apnea?

a) Excessive sleepiness
b) Nocturia
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18. Which one of the following sleep disorders is the least common?

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b) central sleep apnea
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19. Which one of the following statements describes narcolepsy?

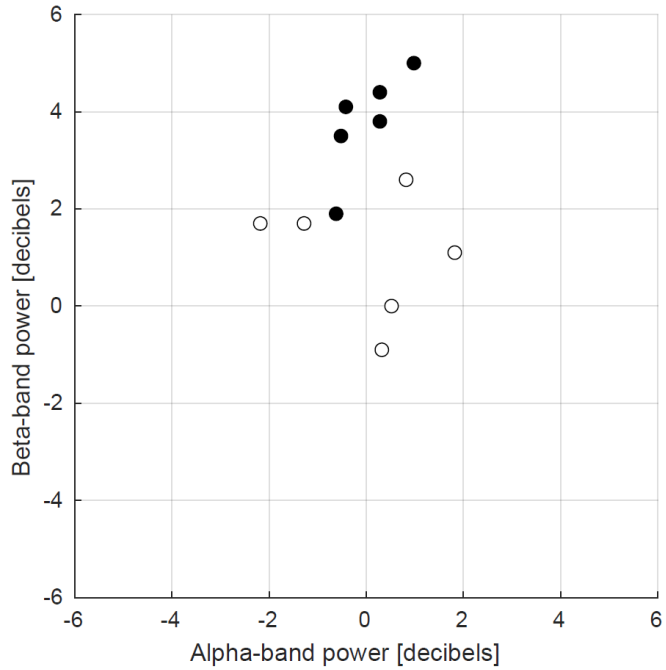
a) It is primarily caused by environmental factors and lifestyle choices.
b) It can be completely cured through behavioural therapy alone.
c) It often involves sudden sleep attacks and may include hypnagogic hallucinations.
d) It is characterized mainly by difficulty falling asleep and staying asleep at night.

20. Under which circumstance would you be least likely to see REM at sleep onset?

a) On withdrawal from REM-suppressing medication.
b) In a newborn human infant.
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21. In the plot below, Student A has represented their electroencephalogram (EEG) recordings in a two-dimensional feature space. There are two samples, each comprising six observations: "sample 1" (open circles) and "sample 2" (closed circles).



These are the observations for sample 1:

0.52	0.00
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-2.18	1.70
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0.28	4.40
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When answering the following questions, use the plot and data above.

i. Are the samples linearly separable? Explain your answer. (2 marks)

ii. Calculate the mean of sample 1 and the mean of sample 2. Clearly mark these means on the plot provided above. (3 marks)

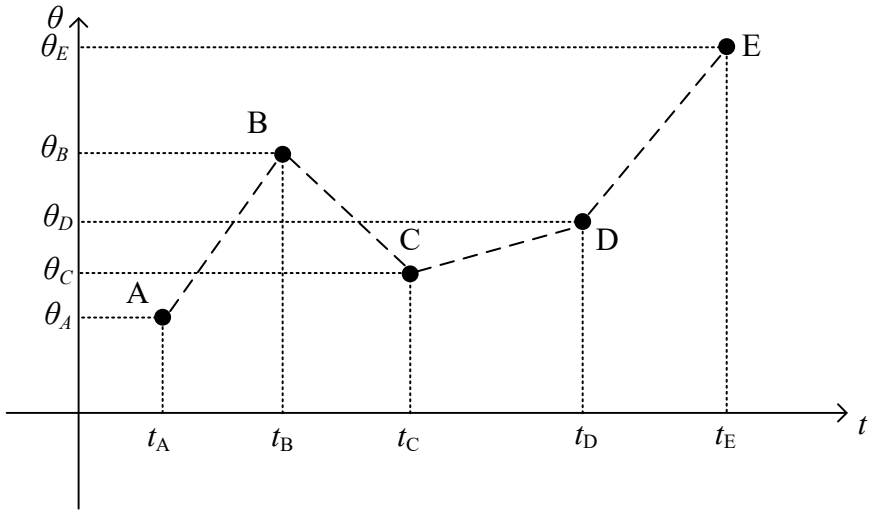
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iii. Using Fisher's linear discriminant, derive an equation for a decision boundary. Mark this decision boundary on the plot provided above. State any assumptions and show your working. (5 marks)

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PART 2: Motion planning, optimal design and biomechanics of skeletal system (40 marks)

1. A robotic joint is at rest at Point A, and performs a series of actions passing through several via points to finally reach Point E and stop there. The positions and corresponding time for each point are known as θ_{A} to θ_{E} and t_{A} to t_{E}, respectively, as displayed in the figure below.



i. State the three rules for motion planning. (3 marks)

ii. Assuming $\theta_{\mathrm{A}} = 10°$, $\theta_{\mathrm{B}} = 30°$, $t_{\mathrm{A}} = 0\ s$ and $t_{\mathrm{B}} = 1\ s$, find the position, velocity and acceleration motion profile for Section AB using cubic polynomial functions. (3 marks)

iii. Supposing the velocities at Point B and C are specified as $\dot{\theta}_B$ and $\dot{\theta}_C$, respectively, give the functions that can be used to find the position, velocity and acceleration motion profiles for Section BC by 5th-order polynomial functions. (6 marks)

iv. Supposing the accelerations at every point are constant and known as $\ddot{\theta}_A$, $\ddot{\theta}_B$, $\ddot{\theta}_C$, $\ddot{\theta}_D$, and $\ddot{\theta}_E$, sketch the velocity profile in a graph for Section CDE using the linear parabolic blend method, and also give all the equations to calculate the durations for accelerating or decelerating phases, suggesting to use t_{D1}, t_{D2} etc for the start and end of the accelerating phase. (Hint: the velocity at Point C is 0, and the velocity at Point D is the average of the linear velocity for Sections CD and DE) (8 marks)

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2. For optimisation problem and genetic algorithms, please answer the following questions.

   i. State the three basic components that define an optimisation problem. (3 marks)

   ii. Using a genetic algorithm to minimise the function of $y=x^2$ with the domain of x-value between 0 and 31, consider a population of four chromosomes, each represented by a 5-digit binary numbers: 11010, 00011, 01010, and 10100. Calculate the selection probability for each chromosome. (3 marks)

   iii. Indicate whether the following statements are **TRUE** or **FALSE.** (2 marks)

   a) The steps of a genetic algorithm are carried out in the sequence of initial selection, evaluation, selection, mutation, and crossover.

   b) The likelihood of a chromosome undergoing crossover is nearly the same as the probability of mutation.

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3. In the context of biomechanics, please answer the below questions.

   i. For a human musculoskeletal system, state the three key elements, and state the corresponding components in the context of a masticatory system. (4 marks)

   ii. For the Hill muscle model, give the expression of the muscular force of a sarcomere, and explain the function of each component of the model. (3 marks)

iii. State the mechanism of the central pattern generator (CPG) control of a musculoskeletal system with the aid of a sketch of the system, and explain the basic function of each component of the system. (5 marks)

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PART 3: Human-robot interaction and medical devices (30 marks)

1. Name three typical requirements for designing a biomechatronics device that physically interacts with a human being. (3 marks)

2. State why variable stiffness and/or damping is a desirable characteristic of a biomechatronics system. (2 marks)

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7. Using what you learnt from the industry guest lectures by Dr Ward, Ms Yee, Prof White and Prof Malpas

    i. Describe one of the examples of a medical device that was presented, and how the device is used to help the patient population it was designed for. (5 marks)

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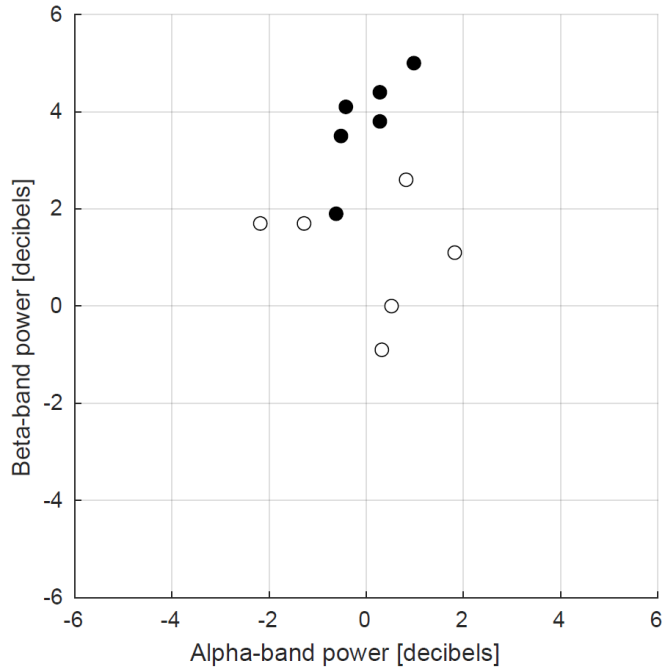
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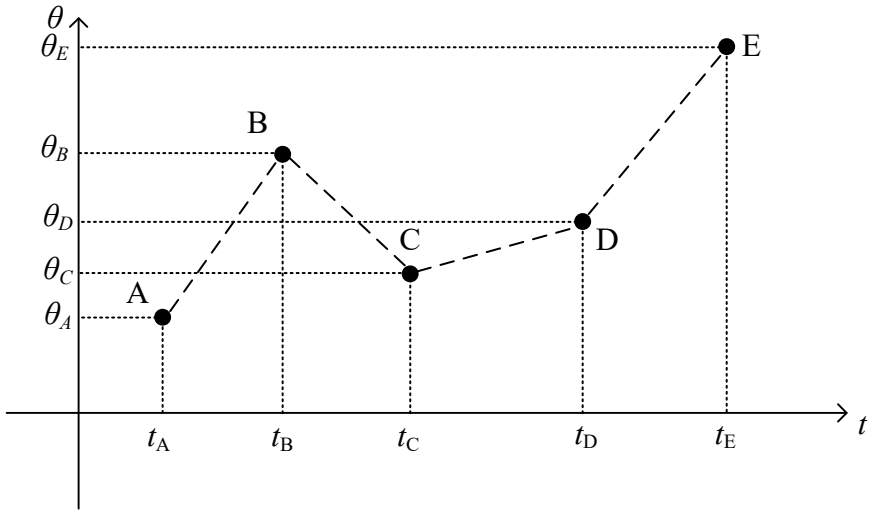
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12. Which one of the following is not a sleep state or a sleep stage?

a) spindles
b) N2
c) non-REM
d) slow-wave sleep

13. Which one of the following electroencephalogram (EEG) patterns is not characteristic of sleep?

a) high-amplitude slow waves
b) alpha rhythms
c) spindles
d) K-complexes

14. Which one of the following statements is incorrect?

a) The traditional terminology for non-REM sleep stages is: stage 1, stage 2, stage 3, stage 4.
b) The new terminology for non-REM sleep stages, introduced by the American Academy of Sleep Medicine in 2007, is: N1, N2, N3.
c) Human slow-wave sleep is synonymous with sleep stage N2.
d) Human slow-wave sleep is synonymous with sleep stage 3 and stage 4.

15. Which one of the following statements is incorrect?

a) Normal sleep begins in non-REM.
b) Normally, the first period of REM sleep occurs about 80 to 100 minutes after sleep onset.
c) REM sleep episodes shorten across the night.
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16. Which one of the following is not a sign/symptom of obstructive sleep apnea?

a) Excessive sleepiness
b) Nocturia
c) Morning headache
d) Hypnogogic jerk

17. Which one of the following statements is correct?

a) Stage N2 sleep constitutes about 13% to 23% of sleep.
b) Stage N3 sleep constitutes about 45% to 55% of sleep.
c) REM sleep constitutes about 12% to 14% of sleep.
d) Stage N1 sleep constitutes about 2% to 5% of sleep.

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18. Which one of the following provides a "shorthand" definition of non-REM sleep?

a) A relatively active brain in a movable body.
b) A relatively inactive brain in a paralyzed body.
c) A relatively inactive brain in a movable body.
d) A relatively active brain in a paralyzed body.

19. Under which circumstance would you be least likely to see REM at sleep onset?

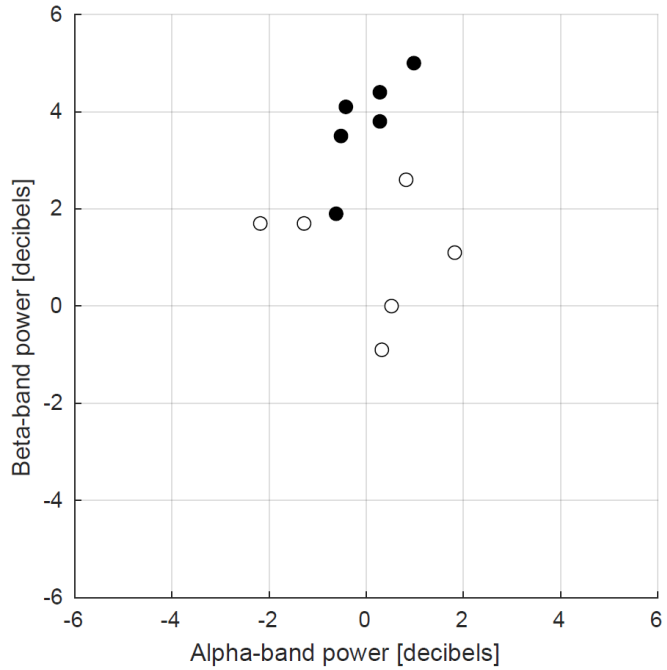
a) On withdrawal from REM-suppressing medication.
b) In a newborn human infant.
c) At bedtime in a healthy young adult.
d) In a patient with narcolepsy.

20. Which one of the following statements about sleep is incorrect?

a) Memory deficits affect events occurring near sleep onset.
b) A sleep clinician should expect to see more slow-wave sleep in older men.
c) Insomnia and hypersomnia are sleep disorders.
d) In normal, young adults there is more non-REM sleep early in the sleep episode as compared to late in the sleep episode.

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21. In the plot below, Student A has represented their electroencephalogram (EEG) recordings in a two-dimensional feature space. There are two samples, each comprising six observations: "sample 1" (open circles) and "sample 2" (closed circles).



These are the observations for sample 1:

0.52	0.00
1.82	1.10
-2.18	1.70
0.82	2.60
0.32	-0.90
-1.28	1.70

These are the observations for sample 2:

0.28	4.40
-0.42	4.10
0.28	3.80
-0.62	1.90
-0.52	3.50
0.98	5.00

When answering the following questions, use the plot and data above.

i. Are the samples linearly separable? Explain your answer. (2 marks)

ii. Calculate the mean of sample 1 and the mean of sample 2. Clearly mark these means on the plot provided above. (3 marks)

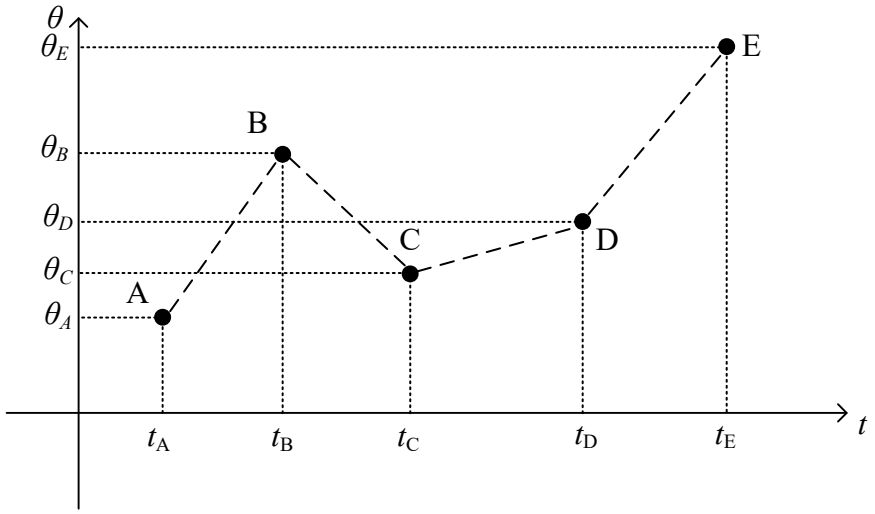
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iii. Using Fisher's linear discriminant, derive an equation for a decision boundary. Mark this decision boundary on the plot provided above. State any assumptions and show your working. (5 marks)

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PART 2: Motion planning, optimal design and biomechanics of skeletal system (40 marks)

1. A robotic joint is at rest at Point A, and performs a series of actions passing through several via points to finally reach Point E and stop there. The positions and corresponding time for each point are known as θ_{A} to θ_{E} and t_{A} to t_{E}, respectively, as displayed in the figure below.



i. State the three rules for motion planning. (3 marks)

ii. Assuming $\theta_{\mathrm{A}} = 10°$, $\theta_{\mathrm{B}} = 30°$, $t_{\mathrm{A}} = 0\ s$ and $t_{\mathrm{B}} = 1\ s$, find the position, velocity and acceleration motion profile for Section AB using cubic polynomial functions. (3 marks)

iii. Supposing the velocities at Point B and C are specified as $\dot{\theta}_B$ and $\dot{\theta}_C$, respectively, give the functions that can be used to find the position, velocity and acceleration motion profiles for Section BC by 5th-order polynomial functions. (6 marks)

iv. Supposing the accelerations at every point are constant and known as $\ddot{\theta}_A$, $\ddot{\theta}_B$, $\ddot{\theta}_C$, $\ddot{\theta}_D$, and $\ddot{\theta}_E$, sketch the velocity profile in a graph for Section CDE using the linear parabolic blend method, and also give all the equations to calculate the durations for accelerating or decelerating phases, suggesting to use t_{D1}, t_{D2} etc for the start and end of the accelerating phase. (Hint: the velocity at Point C is 0, and the velocity at Point D is the average of the linear velocity for Sections CD and DE) (8 marks)

ID: ……

2. For optimisation problem and genetic algorithms, please answer the following questions.

   i. State the three basic components that define an optimisation problem. (3 marks)

   ii. Using a genetic algorithm to minimise the function of $y=x^2$ with the domain of x-value between 0 and 31, consider a population of four chromosomes, each represented by a 5-digit binary numbers: 11010, 00011, 01010, and 10100. Calculate the selection probability for each chromosome. (3 marks)

   iii. Indicate whether the following statements are **TRUE** or **FALSE.** (2 marks)

   a) The steps of a genetic algorithm are carried out in the sequence of initial selection, evaluation, selection, mutation, and crossover.

   b) The likelihood of a chromosome undergoing crossover is nearly the same as the probability of mutation.

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3. In the context of biomechanics, please answer the below questions.

   i. For a human musculoskeletal system, state the three key elements, and state the corresponding components in the context of a masticatory system. (4 marks)

   ii. For the Hill muscle model, give the expression of the muscular force of a sarcomere, and explain the function of each component of the model. (3 marks)

iii. State the mechanism of the central pattern generator (CPG) control of a musculoskeletal system with the aid of a sketch of the system, and explain the basic function of each component of the system. (5 marks)

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PART 3: Human-robot interaction and medical devices (30 marks)

1. Name three typical requirements for designing a biomechatronics device that physically interacts with a human being. (3 marks)

2. State why variable stiffness and/or damping is a desirable characteristic of a biomechatronics system. (2 marks)

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3. Give an example of a variable stiffness actuator and how this could be implemented in a biomechatronics device. (4 marks)

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4. List three types of bio-inspired actuators and an example application for each of the actuators. (6 marks)

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5. Name two types of interaction controller architecture. (2 marks)

6. Briefly describe how an assist-as-needed controller functions to deliver health outcomes for patients undertaking rehabilitation training. (5 marks)

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7. Using what you learnt from the industry guest lectures by Dr Ward, Ms Yee, Prof White and Prof Malpas

    i. Describe one of the examples of a medical device that was presented, and how the device is used to help the patient population it was designed for. (5 marks)

    ii. What are three major challenges to bring a medical technology from research to a developed product that can be used by real patients? (3 marks)

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THE UNIVERSITY OF AUCKLAND

SEMESTER TWO 2024
Campus: City

MECHANICAL ENGINEERING

Biomechatronic Systems

(Time Allowed: TWO hours)

NOTE:

1. This exam is worth 50% of your final grade.
2. It will be marked out of 100 marks.
3. There are 3 parts. Part 1 and 3 are worth 30 marks each and part 2 is worth 40 marks.
4. You must answer ALL questions.
5. Answer each question in the space provided below the question.
6. If you require additional answer pages, overflow pages are provided at the end of this booklet. Within the designated space for the question you answer, you must provide a clear indication that your answer is continued on an overflow page e.g., "Answer is continued on overflow page X".
7. Write neatly and legibly. Use a dark ink that does not show through the paper.
8. Cross out notes and other work you do not wish to have assessed.

Surname: ……..…….. Forenames: ……

University of Auckland ID number: ……

ID: ……

PART 1: EEG and sleep (30 marks)

For each of the following multiple-choice questions (questions 1 to 20), circle one answer only. "REM" refers to "rapid eye movement". Each of these multiple-choice questions is worth 1 mark.

1. Which one of the following provides a "shorthand" definition of non-REM sleep?

a) A relatively active brain in a movable body.
b) A relatively inactive brain in a paralyzed body.
c) A relatively inactive brain in a movable body.
d) A relatively active brain in a paralyzed body.

2. Which one of the following electroencephalogram (EEG) patterns is <u>not</u> characteristic of sleep?

a) high-amplitude slow waves
b) alpha rhythms
c) spindles
d) K-complexes

3. Which one of the following sleep disorders is the least common?

a) obstructive sleep apnea
b) central sleep apnea
c) insomnia
d) restless legs syndrome

4. Which one of the following statements describes narcolepsy?

a) It is primarily caused by environmental factors and lifestyle choices.
b) It can be completely cured through behavioural therapy alone.
c) It often involves sudden sleep attacks and may include hypnagogic hallucinations.
d) It is characterized mainly by difficulty falling asleep and staying asleep at night.

5. Which one of the following definitions best applies to "hypnogram"?

a) A graph that represents the spontaneous electrical activity of the brain.
b) A graph that represents the stages of sleep as a function of time.
c) A graph that represents the heart's electrical activity as patient transitions between sleep and wakefulness.
d) An electrical device used to gather and amplify a biological signal.

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6. Which of the following statements about healthy adult sleep is correct?

a) All of these.
b) The sleep of young adult males shows no difference to that of young adult females.
c) The transition from wake to sleep produces a memory impairment.
d) Sleep episodes comprise ultradian cycles.

7. Regarding sleep onset, which one of the following statements is incorrect?

a) The electromyogram (EMG) may show gradual diminution of muscle tone as sleep approaches.
b) The electrooculogram (EOG) may show slow eye movements as sleep approaches.
c) As sleep approaches, the electroencephalogram (EEG) changes from a low-voltage, mixed-frequency pattern to clear, rhythmic alpha activity.
d) The brain is unresponsive to visual input immediately after sleep onset.

8. Which one of the following statements is correct?

a) Stage N2 sleep constitutes about 13% to 23% of sleep.
b) Stage N3 sleep constitutes about 45% to 55% of sleep.
c) REM sleep constitutes about 12% to 14% of sleep.
d) Stage N1 sleep constitutes about 2% to 5% of sleep.

9. Which of the following is not a risk factor for obstructive sleep apnea?

a) Dehydration
b) High blood pressure (hypertension)
c) Smoking
d) Old age

10. Which one of the following is not a sign/symptom of obstructive sleep apnea?

a) Excessive sleepiness
b) Nocturia
c) Morning headache
d) Hypnogogic jerk

11. Which one of the following statements is incorrect?

a) Normal sleep begins in non-REM.
b) Normally, the first period of REM sleep occurs about 80 to 100 minutes after sleep onset.
c) REM sleep episodes shorten across the night.
d) The REM/non-REM sleep cycle is about 90 minutes in duration.

12. Which one of the following statements about the apnea-hypopnea index (AHI) is incorrect?

a) The AHI is commonly used to define obstructive sleep apnea, and its severity.
b) The AHI is the average number of apnoea and hypopnea events (lasting > 10 sec) that occur every hour.
c) AHI < 5 indicates mild obstructive sleep apnea.
d) AHI > 30 indicates severe obstructive sleep apnea.

13. For people aged 30 to 70 years, what is the prevalence of obstructive sleep apnea?

a) 1%
b) 25%
c) 0.5%
d) Between 70 and 80%

14. Which one of the following statements about sleep is incorrect?

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15. Which one of the following is not a sleep state or a sleep stage?

a) spindles
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16. Which of the following statements about sleep arousal threshold is correct?

a) A person likely has a lower sleep arousal threshold when hearing their own name versus someone else's name.
b) Stage 1 sleep is associated with a high arousal threshold.
c) Children cannot be roused from REM sleep.
d) Low arousal threshold is typical of deep sleep.

17. Under which circumstance would you be least likely to see REM at sleep onset?

a) On withdrawal from REM-suppressing medication.
b) In a newborn human infant.
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18. Which one of the following is the primary purpose of a continuous positive airway pressure (CPAP) machine?

a) To provide medication for insomnia.
b) To improve sleep quality through sound therapy.
c) To monitor sleep patterns.
d) To keep the airway open during sleep.

19. Which one of the following statements is incorrect?

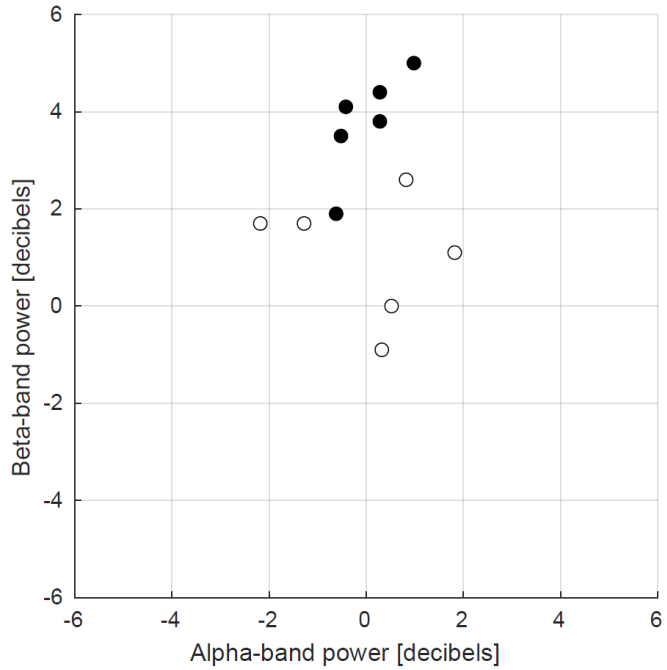
a) Sleep is entered through non-REM sleep.
b) Slow-wave sleep predominates in the last third, and REM sleep predominates in the first third of the night.
c) REM and non-REM sleep alternate with a period near 90 minutes.
d) Wakefulness usually accounts for less than 5% of the night.

20. Which one of the following statements is incorrect?

a) The traditional terminology for non-REM sleep stages is: stage 1, stage 2, stage 3, stage 4.
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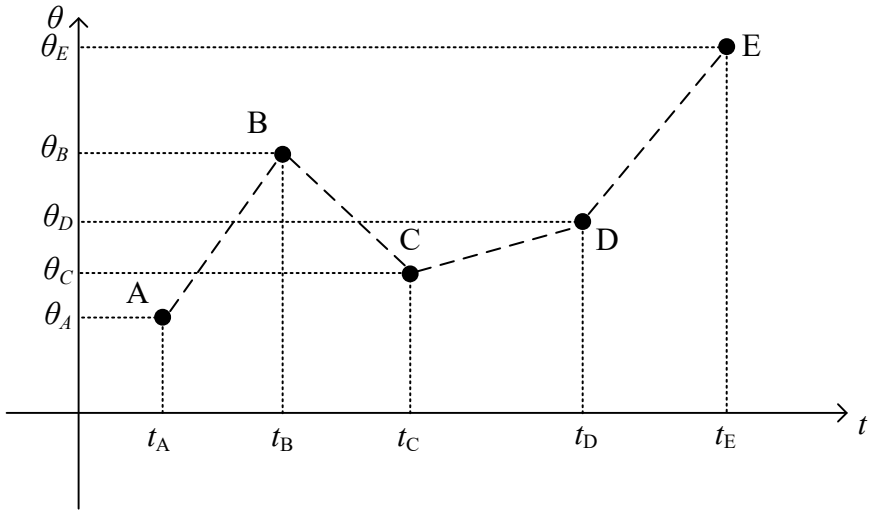
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